

ViPanelTR: A Multi-Agent Framework for Vietnamese Table Question Answering

1st Nguyen Dinh Khoi

University of Information Technology
Vietnam National University Ho Chi Minh City
Ho Chi Minh City, Vietnam
23520774@gm.uit.edu.vn

2nd Vo Tuan Kiet

University of Information Technology
Vietnam National University Ho Chi Minh City
Ho Chi Minh City, Vietnam
kietvt@uit.edu.vn

3rd Dang Van Thin*

University of Information Technology
Vietnam National University Ho Chi Minh City
Ho Chi Minh City, Vietnam
thindv@uit.edu.vn

*Corresponding author: Dang Van Thin (thindv@uit.edu.vn)

Abstract—Table question answering over real-world data remains challenging in zero-shot settings, where models must reason over irregular table structures and handle unanswerable queries. This is particularly evident in Vietnamese benchmarks such as Open-ViTabQA, which contain merged cells, reasoning-intensive questions, and explicit NULL cases. We present ViPanelTR, a new multi-agent inference framework for zero-shot Vietnamese Table QA. ViPanelTR uses five role-conditioned agents for structural, logical, numerical, evidential, and synthesis-oriented reasoning, applies an early answerability gate, and performs bounded self-review, peer deliberation, and answer normalization. Experiments on Open-ViTabQA show consistent improvements over matched single-agent baselines across three backbones, reaching up to 80.66% in terms of F1-score. Additional comparisons with CoAgt and CoQ show that the gains are not only due to using multiple agents, but also to the proposed role design, answerability handling, and normalization strategy.

I. INTRODUCTION

Table question answering over real-world tables requires substantially more than direct cell retrieval. A system must align a natural-language question with latent table schema, interpret irregular layouts such as merged headers or merged value cells, aggregate evidence across multiple rows or columns, and determine when a question is unanswerable because the table does not provide sufficient evidence [1]–[3]. These challenges are especially pronounced in Vietnamese Table QA on Open-ViTabQA [4], which is built from open-domain Wikipedia tables and contains structurally noisy layouts, reasoning-oriented questions, and explicit unanswerable cases.

Prior research on table question answering can be grouped into four main directions. First, semantic parsing approaches map natural-language questions to executable logical forms over semi-structured tables, enabling compositional reasoning but often requiring task-specific supervision and brittle symbolic execution [1], [5]. Second, table-specialized pretraining methods learn table-aware representations directly from

large collections of tabular data, improving cell selection, aggregation, and weakly supervised QA performance [2], [6]–[8]. Third, hybrid table-text models extend QA beyond isolated tables by jointly reasoning over tabular structure and linked passages, which is important when evidence is distributed across tables and surrounding text [9]. Fourth, recent LLM-based methods use prompting, tool use, or multi-step decomposition to perform table reasoning at inference time without heavy task-specific training [10], [11]. Despite this progress, evidence for Vietnamese open-domain table QA remains limited. In particular, prior work has not sufficiently examined whether multi-agent inference-time orchestration can outperform standard zero-shot single-agent prompting on structurally noisy Vietnamese tables while also maintaining reliable abstention for unanswerable questions. This leaves an important gap in both empirical understanding and practical methodology for Vietnamese Table QA.

To address this gap, we propose *ViPanelTR*, a new multi-agent inference framework for Vietnamese Table Question Answering on Open-ViTabQA. ViPanelTR serializes each table into a structured textual representation, decomposes reasoning into five role-specialized agents, incorporates an early answerability gate, and performs bounded self-review, peer deliberation, and answer normalization before prediction. The contributions are threefold: (i) a role-specialized framework whose agents correspond to structural, logical, numerical, evidential, and synthesis needs in Vietnamese table QA; (ii) an answerability-aware pipeline for explicit NULL cases; and (iii) a controlled evaluation with matched single-agent backbones, CoAgt/CoQ comparisons, and fine-grained analyses by question type, answerability, and table structure.

II. RELATED WORK

a) *Vietnamese and open-domain table QA*: Open-ViTabQA [4] targets Vietnamese open-domain table QA over

Wikipedia tables with irregular layouts and explicit unsupported questions. More broadly, table QA spans semantic parsing over semi-structured tables [1], [5], hybrid table-text reasoning [9], and free-form QA [12]. We focus on zero-shot Vietnamese Table QA, where structural grounding and abstention are central.

b) Table reasoning with serialized tables: Table understanding has been improved by table-aware pretraining and fine-tuned readers such as TAPAS, TaBERT, TAPEX, and OmniTab [2], [6]–[8]. Recent work also studies prompting and program-aided reasoning over serialized tables with general-purpose LLMs [3], [10], [11]. ViPanelTR stays in this inference-time setting: it serializes the table and introduces neither a new encoder nor task-specific training.

c) Inference-time orchestration and panel reasoning: Test-time reasoning can be improved through chain-of-thought [13], self-consistency [14], self-refinement [15], and role-conditioned prompting [16]. PanelTR [17] demonstrates the value of staged panel-style discussion for table reasoning, while CoAgt [18] and CoQ [19] organize table reasoning through chain-style agents or SQL-aided collaboration. ViPanelTR is related to this line of work, but differs by targeting Vietnamese open-domain table QA and by integrating answerability gating, answer normalization, and role definitions tied to Open-ViTabQA error modes such as merged cells, unsupported questions, Yes/No answers, lists, and surface-form variation.

III. METHODOLOGY

In this section, we formalize zero-shot Vietnamese Table QA on Open-ViTabQA and describe the ViPanelTR inference pipeline. The method operates entirely at inference time: it converts each table into a structure-aware textual representation, elicits complementary analyses from multiple role-specialized agents, and then reconciles these analyses through an early answerability gate, self-review, peer deliberation, and answer normalization. Figure 1 illustrates the overall pipeline.

A. Problem Formulation

Let $\mathcal{D} = \{(T_i, q_i, y_i)\}_{i=1}^N$ denote a dataset of Vietnamese table-QA instances, where T_i is a table, q_i is a question, and $y_i \in \mathcal{Y}_{\text{ans}} \cup \{\text{NULL}\}$ is the gold output. Here, $\mathcal{Y}_{\text{ans}}$ denotes the set of grounded textual answers supported by the table, while NULL denotes that the table does not contain sufficient evidence to answer the question.

Given an input pair (T, q) , the goal is to predict an answer grounded only in the evidence contained in T . Formally, the system implements an inference function f such that $\hat{y} = f(T, q)$ and $\hat{y} \in \mathcal{Y}_{\text{ans}} \cup \{\text{NULL}\}$. In Open-ViTabQA, this mapping is challenging because a valid prediction may require structural interpretation, multi-condition filtering, comparison, arithmetic reasoning, list aggregation, or faithful abstention [4].

B. Input Representation

Because Open-ViTabQA tables are collected from Wikipedia and often exhibit non-flat structures, we first

transform each raw table T into a textual representation $\tilde{T} = g(T)$. The serialization function $g(\cdot)$ preserves header content, row-column order, and visible span structure, including information induced by rowspan and colspan. We additionally apply Unicode normalization and whitespace cleanup to obtain a stable representation across all backbones.

The final model input is constructed as $S = [\pi; \tilde{T}; q]$, where π is a task instruction shared across agents, $\tilde{T}$ is the serialized table, and q is the Vietnamese question. This design keeps the method compatible with general-purpose LLMs while retaining enough structural information for reasoning over irregular tables [3].

C. Panel Investigation

Open-ViTabQA questions require different capabilities: some ask for schema interpretation over merged cells, some require filtering multiple conditions, some involve comparison or arithmetic, and others mainly require evidence checking before returning NULL. ViPanelTR therefore does not rely on a single general prompt. Given the shared input S , we launch five prompt-conditioned investigations using the same backbone model but different role instructions: Structuralist, Logician, Calculator, Verifier, and Synthesizer [16], [17]. Let $\mathcal{A} = \{\rho_1, \dots, \rho_5\}$ denote this fixed role set. The Structuralist focuses on schema interpretation, header hierarchy, row-column alignment, and merged cells; the Logician handles conditions, negation, and multi-condition reasoning; the Calculator handles arithmetic, comparison, counting, ranking, and numerical units; the Verifier prioritizes evidence grounding and abstention; and the Synthesizer reconciles candidate answers and disagreements.

For each role $\rho_i \in \mathcal{A}$, the initial investigation produces $z_i^{(0)} = \mathcal{M}_{\rho_i}(S) = (r_i^{(0)}, e_i^{(0)}, c_i^{(0)}, u_i^{(0)})$, where $r_i^{(0)}$ is the draft answer, $e_i^{(0)}$ is the cited evidence, $c_i^{(0)}$ is a confidence signal, and $u_i^{(0)} \in \{0, 1\}$ is a binary unsupported judgment. The objective of this stage is not to create learned experts, but to induce diverse inference traces from the same backbone at test time.

D. Early Answerability Gate

Open-ViTabQA explicitly contains unsupported questions, so NULL should not only be treated as a final textual answer competing with fluent but unsupported predictions [4]. We therefore collect the unsupported judgments from the five initial investigations and compute $U^{(0)} = \sum_{i=1}^5 u_i^{(0)}$. If $U^{(0)} \geq 3$, the system directly returns NULL.

This majority-based gate serves two purposes. First, it reduces the tendency of the model to force unsupported questions into spurious textual answers. Second, it prevents later self-review and peer deliberation from rationalizing low-quality hypotheses when the table does not actually ground a valid response. It also reduces inference calls for cases that stop at this stage.

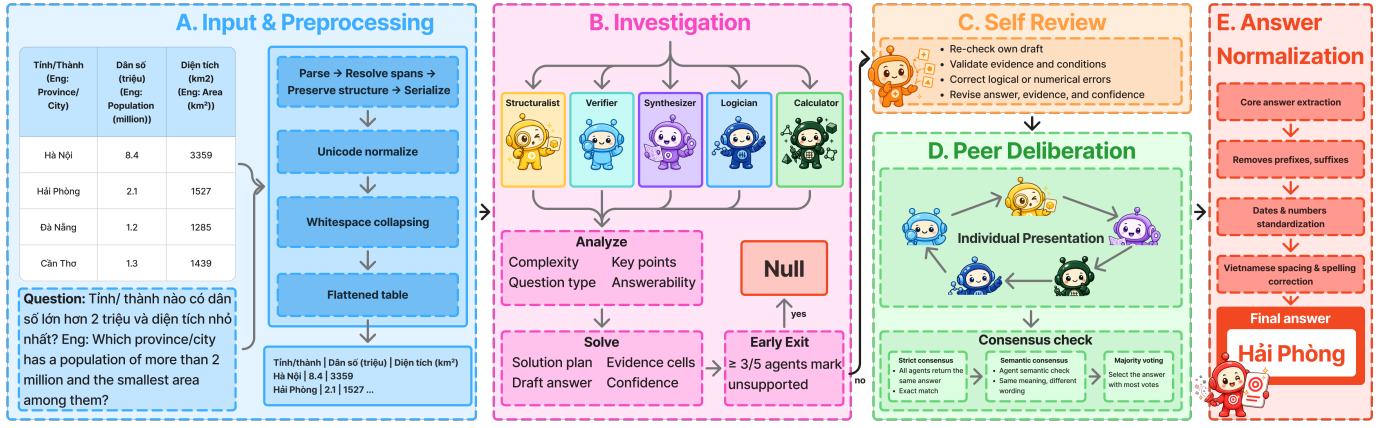


Fig. 1. ViPanelTR pipeline for zero-shot Vietnamese Table QA. The table is serialized, five role-specialized investigations run in parallel, and the system applies an early answerability gate, self-review, peer deliberation, and answer normalization.

E. Self-Review and Peer Deliberation

If the early answerability gate does not trigger, each role performs one bounded self-review step to reassess its own draft in light of the serialized table and the original question. Formally, the revised state is $z_i^{(1)} = \mathcal{R}_{\rho_i}(z_i^{(0)}, \tilde{T}, q)$. This stage is designed to correct local inconsistencies such as unsupported evidence selection, numerical slips, or incomplete condition matching.

After self-review, the five revised responses enter one peer-deliberation round following the staged design of PanelTR [17]. During this stage, the agents compare answers, supporting evidence, and confidence signals. The system first checks strict consensus, then semantic consensus, and otherwise falls back to majority voting with deterministic tie-breaking. The aggregated candidate is $r^* = \text{Vote}(r_i^{(1)}_{i=1}^5)$. We use exactly one self-review round and one peer-deliberation round to keep inference lightweight while still allowing correction and reconciliation across complementary views.

F. Answer Normalization

The final stage converts the aggregated candidate into the surface form returned by the system. Specifically, the prediction is $\hat{y} = \eta(r^*)$, where $\eta(\cdot)$ denotes a lightweight normalization function. This component is important for Vietnamese Table QA because semantically equivalent answers may differ in Yes/No wording, list separators, units, whitespace, or short explanatory phrasing, while EM and F1 are sensitive to surface form. The normalizer canonicalizes such variants and standardizes the rendering of NULL, but it is not allowed to add content unsupported by the table.

IV. EXPERIMENTAL SETUP

A. Dataset and Evaluation Metrics

We evaluate on the official Open-ViTabQA benchmark [4] and report results on its official test set. Following the benchmark protocol, all systems take only the serialized table $\tilde{T}$ and the Vietnamese question q as input, while question-type and answerability annotations are reserved for post-hoc

analysis. We report the four standard metrics used in prior work: F1, EM, ROUGE-1 (R1), and METEOR (MET) [4].

B. Baseline Comparison

We compare our method against five groups of baselines, following the presentation style of prior benchmark papers while keeping the proposed system separate. Our main claim is based on matched within-backbone comparisons between single-agent prompting and ViPanelTR, and on same-backbone comparisons with reimplemented multi-agent systems.

a) *Prompt-based LLMs*: This group includes prompting-based large language models reported in Open-ViTabQA [4], namely Gemini 2.0 Flash Exp., Gemini 1.5 Pro, Mistral Nemo 2407, Mistral 7B v0.3, and LLaMA 2. We also include our three controlled zero-shot prompt-based baselines, namely Qwen3 8B, LLaMA 3.1 8B, and GPT-4o mini, where each backbone directly predicts an answer from the serialized table and question without panel deliberation. These matched single-agent rows serve as the direct prompt-based counterparts to the proposed system.

b) *Fine-tuned Baselines*: We further compare against the fine-tuned baselines reported in Open-ViTabQA [4], including ViT5-Base, ViT5-Large, TAPAS-Base, TAPAS-Large, and the fine-tuned variants of Mistral Nemo 2407, Mistral 7B v0.3, LLaMA 3.1, and LLaMA 2. Taken together, these models provide supervised reference points spanning both table-specialized pretrained architectures and fine-tuned general-purpose LLMs for Vietnamese table QA.

c) *Proposed system (ViPanelTR)*: We instantiate ViPanelTR on the same three backbones: Qwen3 8B, LLaMA 3.1 8B, and GPT-4o mini. This matched setup isolates the effect of the proposed multi-agent framework relative to the corresponding single-agent prompt-based baseline.

d) *Multi-agent systems*: We also reimplement CoAgt [18] and CoQ [19] on Open-ViTabQA using GPT-4o mini. These rows provide same-backbone context for whether the gains come merely from using multiple agents or from the specific ViPanelTR design.

TABLE I
OVERALL PERFORMANCE ON THE OPEN-VITABQA TEST SET.

Category	Model	F1	EM	R1	MET
Prompt-based	LLaMA 2	4.20	6.00	8.80	5.80
	Mistral 7B v0.3	34.90	35.00	44.80	32.80
	Mistral Nemo 2407	35.50	35.60	45.20	33.50
	LLaMA 3.1 8B	56.44	35.38	45.54	42.89
	Gemini 1.5 Pro	59.80	60.80	71.90	49.80
	Gemini 2.0 Flash Exp.	60.50	60.20	70.10	50.00
	GPT-4o mini	64.57	42.44	55.64	53.17
	Qwen3 8B	62.58	40.46	56.62	51.98
Fine-tuned	LLaMA 2	5.69	7.85	9.58	6.42
	TAPAS-Base	30.62	30.29	41.10	28.27
	TAPAS-Large	31.56	31.44	40.02	27.96
	LLaMA 3.1	37.73	36.89	51.17	38.54
	Mistral 7B v0.3	41.28	41.83	50.76	37.08
	ViT5-Base	42.35	42.24	50.35	36.81
	Mistral Nemo 2407	43.11	42.14	53.71	40.38
	ViT5-Large	45.22	45.13	51.87	37.12
Non-LLM baselines	RGCN-RCI	18.12	17.70	23.24	16.71
	KorWikiTQ	46.00	46.00	52.90	37.80
Multi-agent systems	CoAgt	68.87	32.36	45.32	46.89
	CoQ	73.20	57.86	66.36	67.20
ViPanelTR (Ours)	LLaMA 3.1 8B	62.23	40.02	50.92	48.00
	Qwen3 8B	80.06	64.72	74.94	72.12
	GPT-4o mini	80.66	65.52	76.01	73.36

e) Non-LLM baselines: Finally, we retain two additional non-LLM baselines from prior work, namely RGCN-RCI and KorWikiTQ [4], to provide further reference points from earlier table-QA approaches.

V. RESULTS AND ANALYSIS

A. Overall Results

Table I summarizes the main test results. Following the category layout used in prior papers, we group all single-agent prompting systems under the prompt-based category and keep multi-agent systems and ViPanelTR separate. The key comparisons are the matched within-backbone contrast between the prompt-based rows and ViPanelTR, and the same-backbone contrast with CoAgt and CoQ under GPT-4o mini.

The proposed system improves all three matched backbones, but the gain magnitude is clearly backbone-dependent. Relative to their corresponding prompt-based baselines, the proposed system raises GPT-4o mini from 64.57 to 80.66 F1 (+16.09), Qwen3 8B from 62.58 to 80.06 F1 (+17.48), and LLaMA 3.1 8B from 56.44 to 62.23 F1 (+5.79). The EM gains are also large for GPT-4o mini (+23.08) and Qwen3 8B (+24.26), showing that the benefit is not limited to partial token overlap: ViPanelTR also produces more exact final answers. This pattern is important for table QA, where a system can appear reasonable under F1 while still missing the exact cell span, numeric form, or NULL decision.

A second observation is that ViPanelTR changes the relative behavior of backbones. GPT-4o mini is the strongest prompt-based baseline among the matched backbones and remains the

best proposed-system variant, while Qwen3 8B receives the largest F1 gain. This suggests that ViPanelTR does not simply amplify the single-agent baseline; rather, its gains depend on how well a backbone can exploit role-specialized investigation, revision, and aggregation.

The same-backbone comparison with other multi-agent systems further supports the design choice. Under GPT-4o mini, ViPanelTR outperforms CoAgt by +11.79 F1, +33.16 EM, +30.69 R1, and +26.47 MET. It also outperforms CoQ, the stronger of the two reimplemented systems, by +7.46 F1, +7.66 EM, +9.65 R1, and +6.16 MET. These margins indicate that the gains are not explained only by increasing the number of agents; they also depend on how roles, answerability gating, deliberation, and answer normalization are organized.

B. Analysis by Question Type

Table II shows where ViPanelTR helps most relative to the single-agent baseline. The largest and most consistent gains concentrate on question types that require explicit decision making or multi-step grounding: *Yes/No*, *Calculate*, and *Multi-conditions*. ROUGE-1 on *Yes/No* improves by +60.32 for GPT-4o mini, +59.53 for Qwen3 8B, and +31.74 for LLaMA 3.1 8B. The corresponding gains are +14.61, +22.69, and +12.11 on *Calculate*, and +29.28, +27.99, and +13.74 on *Multi-conditions*. These are precisely the cases where a single-pass answer is often brittle: the model must reconcile several constraints, verify arithmetic, or decide whether the evidence supports a binary judgment. The gains therefore align with the five-role design.

The same table also shows that the framework is not uniformly beneficial. *Why* remains unstable across all three backbones, and *How* is still difficult for GPT-4o mini and LLaMA 3.1 8B. More broadly, LLaMA 3.1 8B exhibits mixed behavior, with gains on *Yes/No*, *Calculate*, and *Multi-conditions*, but declines on several factoid or explanatory categories. This indicates that the current discussion protocol is strongest when answers can be grounded through explicit table comparison and voting, but less stable for open-ended formulations whose surface realization is more variable. In other words, ViPanelTR is most reliable when the problem is “which table-supported answer should be chosen?” rather than “how should this explanation be phrased?”

C. Analysis by Answerability

This analysis examines whether the system can identify cases where the table does not provide enough information to answer the question. Examples with a reference answer of NULL are treated as *Unanswerable*, while all remaining examples are treated as *Answerable*; Figure 2 therefore reports class-wise F1 rather than answer-overlap F1.

For the *Answerable* subset, ViPanelTR improves every backbone. Qwen3 8B is already very strong in the single-agent setting (94.60 F1), so the framework adds only a marginal improvement to 94.97. In contrast, GPT-4o mini benefits much more strongly, rising from 82.38 to 94.54, while LLaMA 3.1 8B improves from 86.44 to 91.40. This suggests that, beyond

TABLE II
ROUGE-1 BY QUESTION TYPE ON THE OPEN-VITABQA TEST SET.

Question type	LLaMA 3.1 8B		Qwen3 8B		GPT-4o mini	
	Prompt-based	ViPanelTR	Prompt-based	ViPanelTR	Prompt-based	ViPanelTR
<i>Who</i>	63.75	44.39 (-19.36)	64.01	68.04 (+4.03)	72.35	72.10 (-0.25)
<i>What</i>	68.97	57.83 (-11.14)	76.85	81.97 (+5.12)	72.48	80.36 (+7.88)
<i>When</i>	69.06	67.24 (-1.82)	79.90	84.56 (+4.66)	76.44	82.83 (+6.39)
<i>Where</i>	71.33	63.61 (-7.72)	79.04	80.99 (+1.95)	70.88	78.91 (+8.03)
<i>How</i>	44.17	23.91 (-20.26)	32.65	47.27 (+14.62)	74.21	55.27 (-18.94)
<i>Why</i>	58.26	36.28 (-21.98)	73.96	53.36 (-20.60)	62.59	57.63 (-4.96)
<i>Yes/No</i>	14.29	46.03 (+31.74)	20.63	80.16 (+59.53)	23.81	84.13 (+60.32)
<i>List</i>	44.52	36.02 (-8.50)	75.55	73.71 (-1.84)	66.50	75.11 (+8.61)
<i>Calculate</i>	27.50	39.61 (+12.11)	38.63	61.32 (+22.69)	54.12	68.73 (+14.61)
<i>Multi-conditions</i>	37.58	51.32 (+13.74)	47.39	75.38 (+27.99)	45.04	74.32 (+29.28)

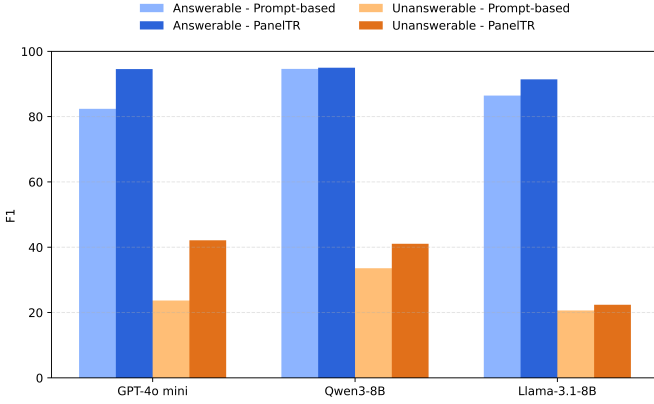


Fig. 2. F1 by answerability on the Open-ViTabQA test set.

TABLE III
PERFORMANCE BY TABLE STRUCTURE ON THE OPEN-VITABQA TEST SET.

Metric	Table structure	LLaMA 3.1 8B		Qwen3 8B		GPT-4o mini	
		Prompt-based	ViPanelTR	Prompt-based	ViPanelTR	Prompt-based	ViPanelTR
F1	Normal	59.24	64.84 (+5.60)	66.09	84.25 (+18.16)	67.13	85.71 (+18.58)
	Header merge	52.69	60.57 (+7.88)	58.51	77.31 (+18.80)	64.11	77.86 (+13.75)
	Value merge	53.16	59.20 (+6.04)	60.27	75.77 (+15.50)	59.57	75.67 (+16.10)
EM	Normal	41.65	46.64 (+4.99)	47.29	70.07 (+22.78)	46.20	72.89 (+26.69)
	Header merge	31.23	36.43 (+5.20)	35.82	62.83 (+27.01)	47.21	64.31 (+17.10)
	Value merge	28.61	32.47 (+3.86)	34.88	58.51 (+23.63)	34.79	57.22 (+22.43)

abstention, ViPanelTR also helps correct answer formulation and evidence selection on genuinely answerable questions.

The harder case is the *Unanswerable* subset. Here the gain is consistent across all backbones: GPT-4o mini improves from 23.66 to 42.11 F1, Qwen3 8B from 33.56 to 41.03, and LLaMA 3.1 8B from 20.62 to 22.37. This result supports the early answerability stage and the later cross-agent reconciliation, since the improvement is concentrated exactly where unsupported-answer detection matters. However, the absolute scores remain far below the answerable subset for all models. Thus, faithful abstention is still the main unresolved challenge in Open-ViTabQA: ViPanelTR reduces unsupported answering, but it does not yet close the gap between extracting evidence and confidently concluding that no grounded answer exists.

TABLE IV
OVERALL PHASE-REMOVAL ABLATION RESULTS ON QWEN3 8B.

Configuration	F1	EM	R1	MET
Inv.+Self+Peer	80.06	64.72	74.94	72.12
Inv.+Self	79.79	64.03	74.69	71.55
Inv.+Peer	79.50	64.72	73.41	71.12

D. Analysis by Table Structure

Table III reports performance by table structure. The main pattern is that ViPanelTR helps not only on regular tables, but also on structurally irregular ones. For Qwen3 8B, F1 improves by +18.16 on normal tables, +18.80 on merged-header tables, and +15.50 on merged-value tables. For GPT-4o mini, the corresponding gains are +18.58, +13.75, and +16.10. Even LLaMA 3.1 8B, although weaker overall, improves across all three structures. This is an encouraging result for the role design: the gains persist even when the model must reconstruct latent schema or trace evidence across non-flat layouts.

EM also increases strongly for Qwen3 8B and GPT-4o mini. Qwen3 8B gains +22.78, +27.01, and +23.63 points across the three structures, while GPT-4o mini gains +26.69, +17.10, and +22.43 points. At the same time, the absolute numbers show that merged structures remain harder than normal ones: for the two strongest ViPanelTR variants, F1 on normal tables reaches 84.25–85.71, but drops to roughly 75–78 on merged-header or merged-value tables. Therefore, ViPanelTR mitigates structural difficulty, but does not eliminate it.

E. Ablation Study

Table IV analyzes the contribution of the refinement stages on Qwen3 8B. The full pipeline (*Inv.+Self+Peer*) gives the best overall F1, ROUGE-1, and METEOR, and ties for the best EM. Removing either self-review or peer deliberation only causes a modest drop, which indicates that the initial multi-role investigation already carries much of the signal. Nevertheless, the full system is consistently the most balanced configuration, so the later phases are not redundant.

The partial variants suggest a clear trade-off: self-review slightly improves answer completeness, while peer deliberation is more useful for conservative rejection of unsupported hypotheses.

TABLE V
EM ABLATION RESULTS FOR ANSWER NORMALIZATION.

Model	With	Without	Delta
Qwen3 8B	64.72	58.17	+6.55
LLaMA 3.1 8B	40.02	38.90	+1.12
GPT-4o mini	65.52	54.13	+11.39

Answer normalization is also a substantial contributor rather than a cosmetic post-processing step. Table V shows that it improves EM for all three backbones, most strongly for GPT-4o mini (+11.39) and Qwen3 8B (+6.55). These gains mainly come from removing unnecessary explanations, canonicalizing Yes/No variants, and formatting lists consistently, while the normalizer is not intended to add content unsupported by the table.

F. Additional Analysis: Answerability Gate Threshold

We also study the threshold in the early answerability gate on Qwen3 8B. The results reflect a trade-off between continuing to produce a textual answer and returning NULL early. When the gate is removed, answerable-class F1 is highest at 82.02, but unanswerable-class F1 drops sharply to 48.82. A threshold of ≥ 4 partially restores unsupported-question detection, reaching 64.33 F1 on the unanswerable class, but remains clearly below the ≥ 3 setting at 71.05. Thus, the majority threshold ≥ 3 gives the best balance in our setting: compared with no gate, it sacrifices only 1.61 points on the answerable class while improving the unanswerable class by 22.23 points.

VI. CONCLUSION AND FUTURE WORK

We presented *ViPanelTR*, a new multi-agent inference framework for Vietnamese table question answering on OpenViTabQA [4]. Under matched-backbone comparisons, *ViPanelTR* consistently outperforms single-agent baselines, with the largest gains on GPT-4o mini and Qwen3 8B, particularly for reasoning-intensive question types such as *Yes/No*, *Calculate*, and *Multi-conditions*. The comparisons with CoAgt and CoQ further indicate that structured role design, early answerability gating, peer deliberation, and answer normalization improve reliability within the same backbone, especially for unsupported-question handling.

Despite these improvements, several challenges remain. Performance on unanswerable questions still lags behind answer extraction, and tables with merged headers or values remain more difficult than regular structures. In future work, we aim to improve answerability calibration, develop structure-aware representations for non-flat tables, conduct role-level ablations, and design more efficient inference strategies that reduce computational cost while preserving the benefits of multi-agent reasoning.

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